Stacks last lecture 'Infix Prefix Postfix'

Queues

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5. Questions on Queues

What is Queue

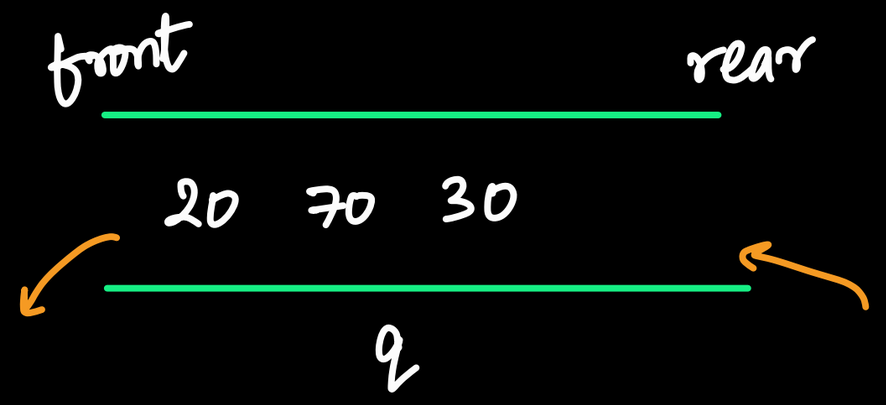
Movie Ticket Counter

Bishal
Preet
Arpit
Santosh
Vineet
Aditya
Ashirwad

FIFO → first in first out

add(val), remove(), peek(), size()

O(1) O(1) O(1)



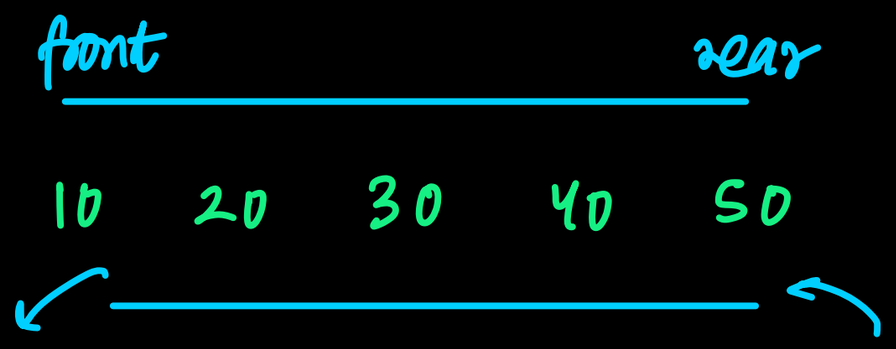

```
q.add(10)
q.add(20)
q.add(70)
sout(q.peek()); // 10
q.remove();
q.add(30)
```

10

add hota hai rear me

remove & peek front me

Traversing a Queue




```
int n = q.size();
for(int i=1; i<=5; i++){
    sout(q.peek());
    q.add(q.remove());
}
```

10 20 30 40 50

add(val), remove(), peek() at any position

is mazedaar in a queue

front rear

10 20 30 60 40 50

q.add(60, 3)

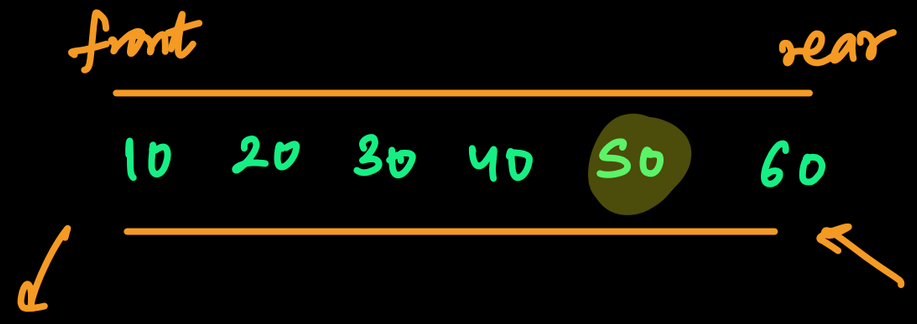
index

1) 3 elements remove & add

2) add

3) n-3 elements remove & add

add(val), remove(), peek() at any position



peek(4)
↓
index

1) 4 elements remove & add

2) peek

3) n-4 elements remove & add

add(val), remove(), peek() at any position

front rear

10 20 40 50

q.remove(2)

1) 2 times remove & add

2) remove

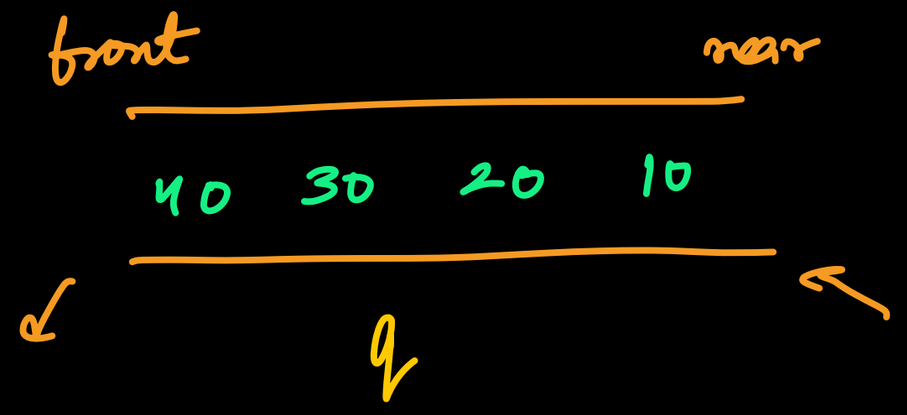
3) q.size()-2 times remove & add

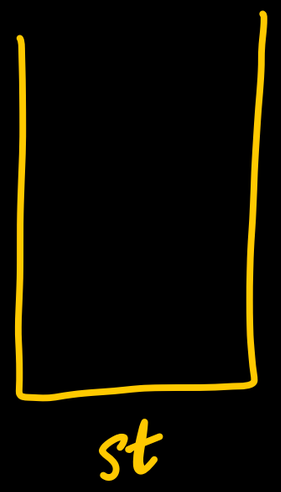
30

Ques: Reverse a Queue

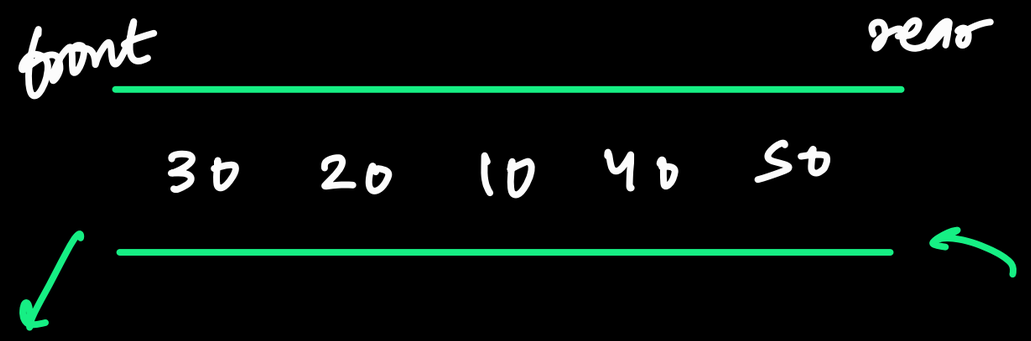
A.S. = O(n)

T.C. = O(n)





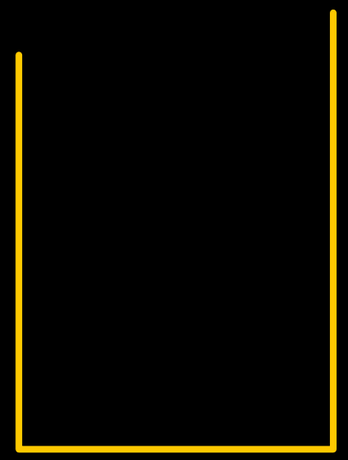
Ques: Reverse first k elements in a queue



$k = 3$

$A.S. = O(k)$

$T.C. = O(n)$



GFG Practice

Ques: Find the Winner of Circular game

11 Arushi
1 Biplab
2 Nikita
3 Riya
4 Hemant
5 Shubhika
6 Arsh
7 Amir
8 Arish
9 Anuj
10 Isha

$n = 11$
$K = 4$

4 8 12 16

9

Leetcode 1823

Ques: Find the Winner of Circular game

$n = 9$
$k = 9$

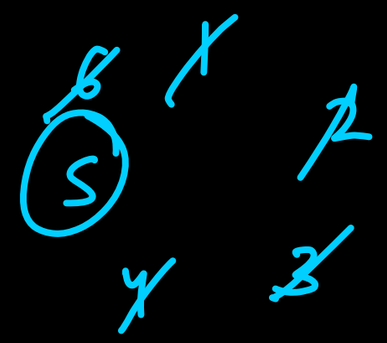
T.C. $= O(k*n)$
A.S. $= O(n)$

7 8 2 4 5

9 , 3 6

Leetcode 1823

Ques: Find the Winner of Circular game



$n = 6$

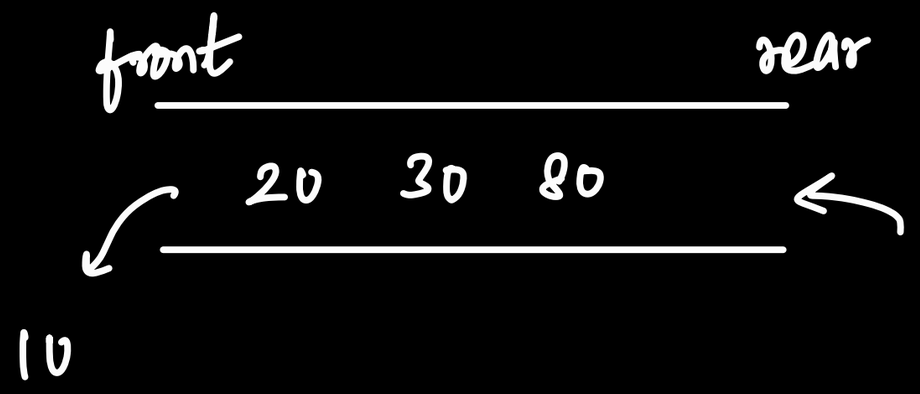
$k = 7$

$n = 1$

$k = ?$

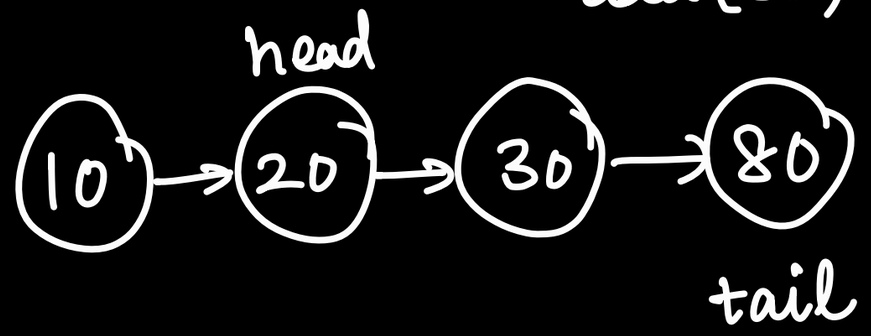
Leetcode 1823

Queue Implementation via Linked List



add(10)
add(20)
add(30)
remove()
add(80)

add
remove
peek



THANKYOU

Cuties